

Why observational completeness is structurally impossible (and why this matters across physics)

Abstract

A recurring aspiration across physics is *observational completeness*: the idea that, given sufficient precision and scope, empirical observation can fully determine the structure of a physical theory. In this paper we argue that this aspiration is **structurally impossible**, independent of technological limitations. Building on the Theory of Axiomatic Necessity (TNA), we show that any empirically accessible domain ($\mathcal{N}_0$) necessarily presupposes a latent structural domain ($\mathcal{N}_1$) that cannot be fully recovered from observation alone. This incompleteness is not epistemic, but ontological: observation is defined only within a closure that already excludes its own structural support. We demonstrate that this constraint appears uniformly across classical field theory, quantum mechanics, statistical physics, and cosmology, and we argue that several persistent theoretical tensions—underdetermination, fine-tuning pathologies, and interpretative inflation—are symptoms of attempts to enforce an impossible observational closure. The result is a unifying meta-criterion: theories fail not when they lack data, but when they demand observational completeness where structure forbids it.

1. Introduction

Physics has advanced through an extraordinary expansion of observational reach, from telescopes and interferometers to particle colliders and cosmological surveys. Yet alongside this expansion persists a recurring dissatisfaction: multiple inequivalent theories remain compatible with the same body of data, and increasingly elaborate constructs are introduced to preserve observational adequacy.

The standard response frames this as a *temporary epistemic gap*: more data, higher precision, or new probes will eventually close the theory. In this paper we argue that this expectation is mistaken. The problem is not insufficient observation, but a **structural impossibility of observational completeness itself**.

We propose that many foundational tensions in modern physics arise from an unexamined assumption: that the observable domain is, in principle, self-sufficient. The Theory of Axiomatic Necessity (TNA) denies this assumption and provides a formal framework for understanding why it fails.

2. Observational completeness: the hidden axiom

By observational completeness we mean the claim that:

Given a sufficiently rich set of observations, the underlying physical structure can be uniquely reconstructed.

This claim is rarely stated explicitly, but it underwrites:

- parameter inference programs,
- claims of empirical closure,
- and the dismissal of alternative formalisms as “interpretative” rather than physical.

Crucially, observational completeness is not an empirical hypothesis; it is a **meta-axiom** about the relationship between observation and structure. As such, it cannot be validated by observation without circularity.

3. The TNA framework: $\mathcal{N}_0$ and $\mathcal{N}_1$

TNA formalizes a minimal structural distinction:

- $\mathcal{N}_0$: the observable, operationally defined domain;
- $\mathcal{N}_1$: the latent structural domain required for $\mathcal{N}_0$ to be coherent.

The central TNA result is:

No nontrivial system can be ontologically closed at $\mathcal{N}_0$.

That is, the conditions that make observation possible cannot themselves be fully observed within the same domain. Attempting to do so generates either inconsistency or surplus structure.

This result is independent of scale, dynamics, or ontology. It applies wherever observation is defined relationally.

4. Structural incompleteness across physical domains

4.1 Classical field theory

Gauge redundancy illustrates that multiple mathematically distinct configurations correspond to the same observables. Attempts to eliminate this redundancy do not remove the underlying structural dependence; they merely relocate it.

4.2 Quantum mechanics

The quantum state is not uniquely recoverable from measurement statistics without additional structural assumptions. Interpretations proliferate precisely because observation underdetermines structure.

4.3 Statistical physics

Macrostates erase microstructural detail by necessity, not ignorance. No increase in macroscopic observation recovers the lost degrees of freedom.

4.4 Cosmology

Large-scale observations are compatible with radically different global structures. Attempts to enforce closure introduce unobservable entities whose role is to stabilize the observational narrative rather than derive from it.

5. Why more data cannot fix this

A common objection is that incompleteness merely reflects current limits. TNA shows why this fails: increasing observational resolution expands $\mathcal{N}_0$ but simultaneously **pushes $\mathcal{N}_1$ outward**. The boundary does not converge.

This is analogous to Gödelian incompleteness, but structural rather than logical: the act of formalizing observation generates new necessary structure beyond its reach.

6. Consequences for theory construction

Recognizing the impossibility of observational completeness reframes several debates:

- **Underdetermination** is structural, not pathological.
- **Fine-tuning problems** arise when latent structure is misclassified as dynamical coincidence.
- **Interpretative inflation** reflects attempts to close $\mathcal{N}_0$ by narrative rather than structure.

The appropriate response is not to eliminate $\mathcal{N}_1$, but to acknowledge and constrain it axiomatically.

7. A criterion of structural honesty

We propose a minimal meta-criterion:

A theory is structurally honest if it explicitly acknowledges the necessary incompleteness of its observational domain.

Such theories may be incomplete, but they are not incoherent. Conversely, theories that claim observational closure must pay the cost elsewhere—in excess ontology, hidden parameters, or untestable dynamics.

8. Conclusion

Observational completeness is not an attainable goal of physics. It is a category error arising from treating observation as ontologically self-grounding. Through TNA, we have shown that incompleteness is not a failure of science but a structural condition of intelligibility.

This recognition does not weaken physics; it clarifies its limits and redirects effort from impossible closures toward principled structural constraints. Across domains, the choice is not between complete and incomplete theories, but between those that hide their incompleteness and those that make it explicit.

References

- [1] C. Bresciano, *Theory of Axiomatic Necessity (TNA) v1.2*, Zenodo (2025).